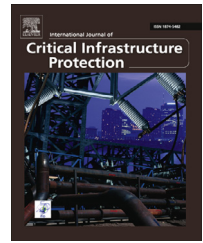


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The challenge of detecting false documents at the border: Exploring the performance of humans, machines and their interaction

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ABSTRACT

The verification of a travel document authenticity is the cornerstone of border checks. Because of pressures at the border (time constraints and increasing numbers of travellers to check) and the growing complexity of modern document security, border control capabilities (officers and/or automated systems) are tasked to rapidly and straightforwardly decide on whether the documents presented are genuine or false. Although operational performance of document inspection capabilities is of paramount importance for the effectiveness and security of border checks, the topic is rarely discussed and studied. The purpose of this article is to contribute to the development of a methodology to assess the operational performance of document inspection capabilities (human and technical) by reporting on the findings of the Document Challenge II, the second exercise of this sort performed in Lisbon in September 2013 and organized by Frontex in cooperation with the Portuguese Serviço de Estrangeiros e Fronteiras (SEF), the UK National Document Fraud Unit (NDFU), the Dutch Royal Marechaussee and the German Forensic Institute of the Bundeskriminalamt (BKA).

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1. Introduction

Document fraud is widely perceived as being a serious problem affecting the borders and the internal security of the Schengen area. The verification of the authenticity of travel documents is a cornerstone of border checks: the information contained in a travel document accepted as genuine is used as a token to verify the identity of the holder and to search watch-lists. A false document accepted as genuine thus invalidates the whole border check process. Because of the sophistication of the physical, optical, and electronic security features of travel documents currently in circulation, border control officers need to increasingly rely on the support of document

inspection systems (DIS) using advanced functionalities and automated components.

Although operational performance of document inspection capabilities is of paramount importance for the effectiveness and security of border checks, it is rarely taken in serious consideration as a vulnerability when assessing the risk of document fraud [1,2]. When it is, no global or European standards and/or structured harmonized methodology exist for the evaluation of the operational performance (accuracy and effectiveness), for the analysis of potential vulnerabilities (errors), and for the design of capacity-building measures of actually deployed capabilities in the operational (theater). When the performance of document inspection capabilities is considered at the operational level, especially in relation to the procurement of technical equipment, important variables

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such as the human factor (including human-machine interaction and usability), the design of travel documents, or the overall decision-making process related to the border check task are not adequately accounted for [3–5]. Finally, empirical data on performance (accuracy and errors) of document inspection capabilities do not exist or are not publicly available so that it is difficult to engage in an evidence-based, rather than normative and theoretical, discussion of vulnerabilities and practical operational and tactical countermeasures.

The purpose of this article is to contribute to the development of a methodology to assess the operational performance of document inspection capabilities (human and technical) by reporting on the findings of the Document Challenge II, the second exercise of this sort performed in Lisbon in September 2013 and organized by Frontex in cooperation with the Portuguese Serviço de Estrangeiros e Fronteiras (SEF), the UK National Document Fraud Unit (NDFU), the Dutch Royal Marechaussee and the German Forensic Institute of the Bundeskriminalamt (BKA). Although the project addressed a variety of related issues (i.e. Human Machine Interaction, usability, document design and forgery analysis), this article focuses only on the evaluation metrics and results of the tests on performance assessment of 7 participating document inspection systems (the “machines”) and 39 border guard experts (the “humans”).

The rest of the article is structured as follows. First, Section 2 provides an overview of related work. Next, Section 3 describes the experiment set up, including details on datasets and evaluation methodology. Subsequently, Section 4 presents evaluation results and an analysis thereof. Finally, Section 5 gives summary conclusions and an outlook as regards future research.

2. Related work

Existing publicly available research in the field of the effectiveness and performance of document fraud detection is scarce. When it exists and is published, it tends to focus on automated systems. [6] and [7] report on the results of tests conducted in the context of the project IDEAL of the Forensic Institute of the German Bundeskriminalamt, focusing on optical functionalities of the DIS. The articles highlight the heterogeneity of the performance of the systems under evaluation and the limitations of machine authentication. Other governmental bodies in Europe (i.e. the Dutch ECID Advies & Innovatie and the UK National Document Fraud Unit) have conducted similar tests the results of which are however restricted. [3] discusses the vulnerabilities of fully automated document checks in an automated border control scenario and highlights the pitfalls in terms of optical and electronic security features detection. Relevant research has also focused on optical benchmarking of document readers [8,9], compression of images [10], document aging and automated security document authentication [11]. The New Technology Working Group of ICAO has published in 2016 Best Practice Guidelines for Machine-Assisted Document Authentication [12].

For what concerns the performance of human document inspection, [13] and [14] show that typically passport control officers make a substantial number of errors on the task of

comparing the photo on the passport and the face of the person in front of them in order to assess whether the document belongs to the person presenting it. Although this indicates a vulnerability in the detection of impostors (a form of identity fraud) using a genuine passport, it does not necessarily address the performance of the task to identify whether the document itself has been tampered with and is false (forged or counterfeited).

The complexity of the task of document authentication at the border and the role of the human factor was the object of a Note presented to the European Union Council Working Party on Frontiers/False Documents by Portugal in November 2010 [15]. The note discussed results of a preliminary study on the reasons why detection of false documents seemed to be declining and on issues related to the performance of automated document examination equipment.

This article builds upon the experiences and results of the national initiatives mentioned above and of the Document Challenge I [16], a preliminary exercise conducted in Amsterdam in June 2012. In particular, it tries to redress the 2012 shortcomings by elaborating a formal metric for performance evaluation and by improving the research protocol in terms of human participants (i.e., a bigger number thereof and with specific first and second line experience), document datasets and testing methodology.

3. Experiment protocol

The Document Challenge II was performed at SEF from 18 September to 1 October 2013. For heuristic reasons, the document inspection task was extrapolated from the overall border check task so that participants (human and machines) were only challenged to correctly classify genuine and false documents in the scenarios with and without time restriction (the parallel verification of other entry requirements was excluded). This artificial de-coupling of the document inspection task from other parallel tasks performed by the border guards when assessing the existence of the conditions for entry (such as identity verification, database check etc.) was necessary in order to eliminate from the classification decision other factors not related to the inspection of the document per se (i.e. passenger profiling). Document inspection or authentication is here understood as the task of comparing a given document against the “ideal-type” (the specimen) of that same document, assessing the degree of similarity between the two and coming to a binary genuine/false decision. The exercise was designed to collect data and observations on the performance of the document authentication task in these three scenarios: *Human only*: Border Control Officers inspecting and assessing the authenticity of a document without the help of advanced technical equipment (only UV light and magnifying glass) with and without time constraint; *Human with machine*: Border Control Officers inspecting and assessing the authenticity of a document with the support of an automated document inspection system and without time constraint; *Machine only*: Automated document inspection systems inspecting and assessing the authenticity of a document alone

Table 1 – Evaluation datasets' statistics: number of genuine/false document, DIF/MED/EAS - respective percentage of difficult, medium and easy documents.

Dataset	genuine	false	DIF	MED	EAS
Test121	53	68	17.4%	24%	58.6%
Test121(95)	48	48	14.6%	24%	61.4%
Test121(25)	5	20	28%	24%	48%

without an officer physically handling the document and without time constraint.

3.1. Dataset

The dataset (which will be referred to with Test121) created for the experiments consists of 121 real genuine and false travel documents detected at UK and Dutch borders, where ca. 78,5% of the documents are passports, 19,8% are ID cards and 1,6% are Visas. Approx. 66% of all documents are EU documents. The dataset was built in such a way as to provide a realistic representation of the travel documents inspected at EU-external/Schengen borders, although the distribution of the different forgeries in the dataset does not necessarily correspond to current trends of document fraud detection. As regards false documents contained in the dataset the forgeries can be subdivided into: (a) counterfeits, which account for 26,7% of all forgeries, (b) documents with substituted images (38,4%), (c) documents with substituted pages (27,9%), and (d) documents with stolen blank (7,4%). Not all of the documents were machine readable travel documents (MRTDs) according to ICAO Doc9303 standard.

Test121 dataset was further subdivided into: (a) Test121(96) – balanced subset consisting of equal number of genuine and false documents, and (b) Test121(25), consisting mainly of false documents. The experts who supplied the documents classified them into three categories in terms of authenticity detection difficulty level (i.e., 'difficult', 'medium' and 'easy'). The statistics of all the three evaluation datasets is provided in Table 1. Test121(25) dataset contains a higher fraction of more 'difficult' documents. It is important to note that neither human testers nor the vendors providing the DIS were informed about the composition of the various datasets wrt. distribution of genuine/false documents and expected difficulty level of authenticity detection.

3.2. Participants

Testers (Humans): Two types of testers participated in the exercise. 2 Master Testers, experienced document experts from the UK and the Netherlands members of the research team, performed the tests in the Machine Only Scenario. 39 European border guards from 15 different Member States took part in the Human Only and Human with Machine scenarios. 20 of them were operational officers deployed in Lisbon in the context of the Joint Operation Meteor (JO Officers), 16 were document expert members of the Frontex Reference Manual Working Group (Reference Manual) [17], while 3 were SEF administrative staff without experience in document or border control.

DIS (Machines): A document inspection system is here defined as a system composed of a scanner and software for optical and electronic authentication of travel documents. Seven document inspection systems from seven different vendors were recruited through two rounds of public call published on Frontex official website in spring 2013¹. Each participating vendor was asked to provide two sets of equipment configured for border check operations (and associated highest security threshold/level). All of the systems except one fulfilled the requirements of the call and were capable of: a) authenticating Passports/BioPage, VISA stickers and EU ID cards; b) generating images in several spectra, from Infrared (IR) light to one or more frequencies of Ultraviolet (UV) light; c) comparing the images taken against a reference security features/document template database; d) reading the electronic chip in ICAO compliant electronic passports; e) presenting the authentication results in an automated and clear way; and f) creating log files of all document inspection processes. Electronic CSCA authenticity checks were not included. Full technical specifications for all of the participating devices were collected. Only 6 of the systems were used for the Machine Only scenario: one was excluded because it did not provide a clear authentication result (it provided only fine-grained analysis of the document). All 7 were used in the Human with Machine scenario.

3.3. Exercise procedure

In the Human Only scenario, each Tester inspected each document of Test121(96) with a time constraint of 15 seconds per document. The result of the inspection in terms of whether the document was classified as genuine, false, or inconclusive was reported using a web form. In the Human with Machine scenario, each Tester inspected a fraction of Test121 using all seven DISs (i.e. a subset of 25 documents on each machine) and reported the results as presented by the machine, whether he/she agreed with the results, and, in case of disagreement, the corrected result. 97.51% of the 7797 document examination records were used for the analysis. In the Machine Only scenario, only the Master Testers inspected the Test121 with 6 DISs and collected the results. Machine logs (8239) of all inspection rounds were saved but could not be used for the analysis because of their heterogeneous format.

3.4. Evaluation

Since the operational aim of border checks is on the one hand to facilitate the crossing of legitimate travellers (bearers of genuine travel documents) and on the other to ensure security by correctly identifying fraudsters and false documents, the evaluation of the performance of document inspection capabilities is broken down into three tasks, namely, evaluation of how good the capability (e.g., humans, machines, or a combination thereof) is at: (a) detection of false documents, (b) detection of genuine documents, and (c) classification of the documents into genuine and false document sets. The evaluation metrics have been broken down into three groups accordingly.

¹ <https://frontex.europa.eu/research/invitations/document-challenge-ii-invitation-to-document-inspection-system-hardware-manufacturers-F9MRgU>.

Table 2 – Metrics for evaluation of Detection of False and Genuine Documents: TP - True Positives (number of genuine documents identified as genuine correctly), TN - True Negatives (number of genuine documents identified as genuine correctly), FP - False Positives (number of false documents identified incorrectly as genuine), FN - False Negatives (number of genuine documents identified incorrectly as false), IP - Inconclusive Positives (number of genuine documents for which an inconclusive answer was returned), IN - Inconclusive Negatives (number of false documents for which an inconclusive answer was returned).

	False document detection	Genuine document detection
Completeness	$TrueNegativeRate(TNR) = \frac{TN}{TN+FP+IN}$	$TruePositiveRate(TPR) = \frac{TP}{TP+FN+IP}$
Error	$FalsePositiveRate(FPR) = \frac{FP}{TN+FP+IN}$	$FalseNegativeRate(FNR) = \frac{FN}{TP+FN+IP}$
Indecision	$NegativeIndecisionRate(NIR) = \frac{IN}{TN+FP+IN}$	$PositiveIndecisionRate(PIR) = \frac{IP}{TP+FN+IP}$
Precision	$NegativePredictiveValue(NPV) = \frac{TN}{TN+FN}$	$PositivePredictiveValue(PPV) = \frac{TP}{TP+FP}$
Overall Performance	$NegativeEffectiveness(NE) = \frac{(1+\beta^2) \cdot (NPV \cdot TNR)}{(NPV \cdot \beta^2) + TNR}$	$PositiveEffectiveness(PE) = \frac{(1+\beta^2) \cdot (PPV \cdot TPR)}{(PPV \cdot \beta^2) + TPR}$

Although each document in our test document collections is either genuine or false, the answer returned by a human or machine can be one of the following: *genuine*, *false* or *inconclusive*. Various reasons for an answer to be considered inconclusive exist. In the case of humans an answer is considered to be inconclusive if she/he is uncertain as regards the authenticity of a document (not sufficient evidence to decide), and explicitly states so. In the case of machines, an answer is considered to be inconclusive if the machine either fails to provide a result (e.g. due to a mechanical failure) or the result returned is unclear and/or difficult to interpret by a human. There are various ways of interpreting inconclusive answers in the evaluation process [18], e.g., excluding them from the evaluation or interpreting the result as one of the underlying classes, i.e., in our scenario, interpreting documents tagged as ‘inconclusive’ as either ‘genuine’ or ‘false’.

For the sake of the evaluation we have either used known metrics or adopted and extended them accordingly. In particular, in order to embrace the inconclusive answers mentioned above, which turned out to occur frequently in our experiments, we introduced a notion of inconclusive positives (negatives) – the number of genuine (false) documents in a test collection for which an inconclusive answer was returned.

As regards the two elementary tasks, i.e., the detection of false and genuine documents, Table 2 provides the metrics used to assess the different aspects of the performance. True Positive Rate (detection of genuine documents) and True Negative Rate (detection of false documents) for measuring “completeness” are also referred to as Sensitivity (Recall) [19,20] and Specificity [21] resp., whereas False Positive Rate and False Negative Rate are also referred to as False Acceptance Rate [21] and False Rejection Rate (False Alarm) [22] resp. For consistency reasons we unified the naming convention through the use of the terms “Positive” and “Negative” in all metrics. The Positive/Negative Indecision rates correspond to the fraction of false (genuine) documents for which an inconclusive answer was returned, whereas Negative/Positive Prediction Value scores capture “precision”. Finally, Negative/Positive Effectiveness metrics are weighted harmonic means² of precision and completeness (recall),

where $\beta = 1$ gives an equal weight to precision and completeness.

Analogously, as regards the document classification task, we have measured Accuracy Rate (ACC), Error Rate (ER) and Indecision Rate (IR), which are defined as follows.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN + IP + IN} \quad (1)$$

$$ER = \frac{FP + FN}{TP + TN + FP + FN + IP + IN} \quad (2)$$

$$IR = \frac{IP + IN}{TP + TN + FP + FN + IP + IN} \quad (3)$$

While the breakdown of the evaluation of the performance of document inspection into three tasks and the definition of the respective measures presented in this section was mainly motivated by the aspects relevant from the operational perspective it is important to emphasize that the problem of detection of false documents could have been evaluated in terms of sensitivity and response bias [23], i.e., a tendency to select one choice over the other.

4. Results

4.1. Humans

Fig 1 provides a histogram reflecting overall human tester performance on Test121(96) dataset in the time-restricted scenario in terms of Accuracy, TPR and TNR which correspond to the document classification into genuine/false ones, detection of genuine documents and detection of false documents resp.

The average accuracy, indecision and error rate for human testers is 65.71%, 20.14%, 14.16% resp. Excluding the scores of the three non-experienced testers, accuracy in the time restricted scenario ranges from a maximum of 94.79% to a minimum of 23.96% with a median of 70.31% and a standard deviation of 15.85%.

TestSet121(25) was used for the evaluation of the human performance in the non-time restricted scenario, from which

² Given the relatively small datasets used in our experiments the effectiveness metrics could have been computed using un-

adjusted means, however, our primary goal was to introduce an evaluation methodology applicable to datasets of any size.

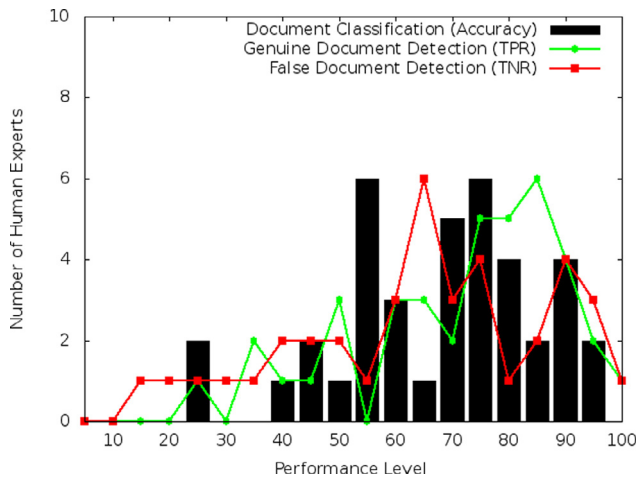


Fig. 1 – Performance Level of Human Experts for Test121(96) in the time-restricted scenario. The values on the y-axis reflect the number of human experts obtaining given performance level.

the non-experienced testers were excluded. Although due to logistics it was not possible to run the exercise with and without time constraint using two comparatively balanced and sized data-set, a quick comparison between the performance of the officers in the time restricted (equal number of false and genuine documents) and non-time restricted scenario (with a majority of false documents) revealed an increase of the median in the classification task to 78% (circa 8% increase) with a standard deviation of 17.09% (max=96%, min=44%, outlier=8%) and with only 8 out of 36 testers experiencing a degradation in classification performance.

High standard deviations in both scenarios indicate that there is high variability in the performance of the testers. Although this was expected because of the idiosyncratic nature of human performance, the extent of the phenomenon highlights the importance of the individual decision-making of the first line officer and the intrinsic vulnerability of first line document inspection (and *ceteris paribus* the entry decision) because of the inevitably subjective assessment of the officer (human factor), thus resulting in an inconsistent level of effectiveness of controls across different officers.

Without time constraint the testers detected correctly more false documents (increase of TNR of 9.79%) and were also more precise (increase of NPV of 11.75%). However, as regards the performance of the task of genuine document detection, the availability of more time for the inspection resulted in a degradation in both completeness (decrease of TPR of -14.58%) and precision (decrease in PPV of -18.63%). This means that an increase of time for inspection improved security (the detection of false documents) with a decrease of 7.5% in false acceptance (less false documents erroneously accepted as genuine for 26 officers out of 36) but at the cost of a considerable increase of false rejection of nearly 20% (more genuine documents erroneously rejected as false for 28 officers out of 36).

Background information on variables such as level, length and type of training received, length and type of operational experience in document inspection and border control, number and types of false documents detected in the past

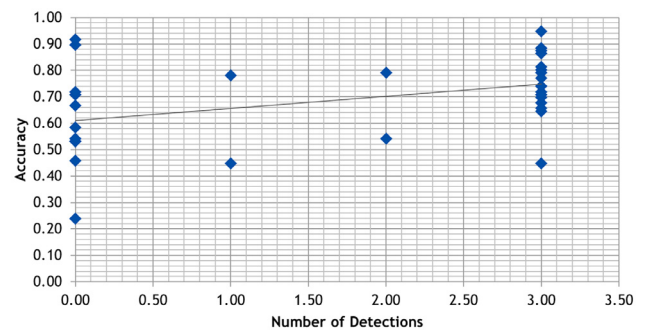


Fig. 2 – Correlation between qualitative operational experience and the performance (accuracy) in the document classification task in the time restricted scenario ($R=0.40$, $p\text{-value} = 0.00$).

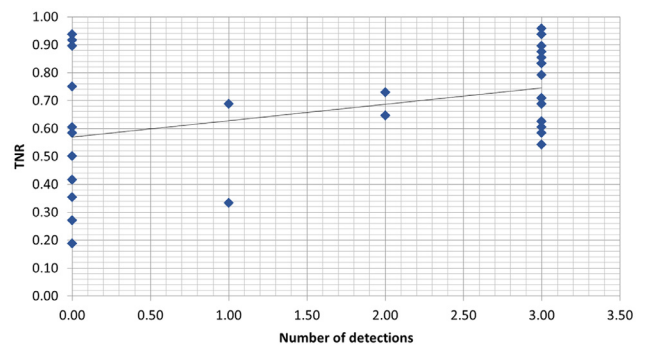


Fig. 3 – Correlation between qualitative operational experience and the performance in the false document detection task in the time restricted scenario ($R=0.41$, $R^2=0.34$, $p\text{-value} = 0.00$).

two years, current position and tasks were collected using a questionnaire before the beginning of the exercise. The intention was to investigate whether a particular factor (training, experience, etc.) could be correlated to performance results. Performance outputs of the testers have been analysed on the basis of the background information available. Due to the fact that the independent variable is just of four levels and the data in the intermediate levels is sparse, correlation is not really informative but could be used in order to engage in a qualitative discussion with the participating officers and better understand their perception related to the causes underlying poor performance. Although no strong correlation exists between performance and any of the variables identified, a moderate positive correlation exists between qualitative operational experience and accuracy in the document classification task (see Fig. 2) in the time restricted scenario (correlation coefficient $R = 0.40$, coefficient of determination $R^2 = 0.33$, statistical significance $p\text{-value} = 0.00$) on the one hand, and qualitative operational experience and the performance in the false documents detection task (see Fig. 3) in the time restricted scenario ($R = 0.41$, $R^2 = 0.34$, $p\text{-value} = 0.00$). Qualitative operational experience has been defined as the number of detections of false documents that the tester has personally performed in his/her operational practice in the past two years. This means that the exposure and the

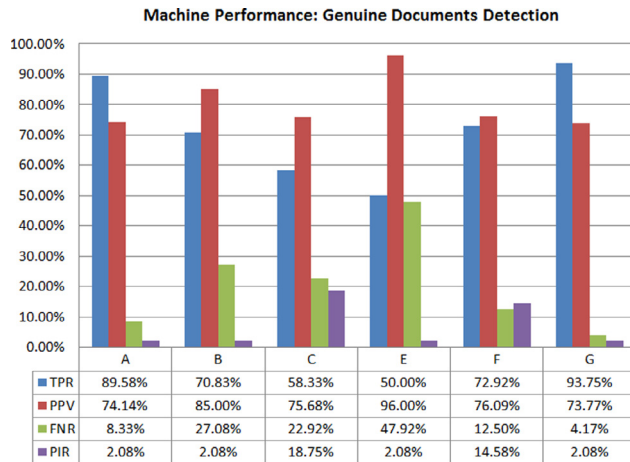


Fig. 4 – Machine Performance on Test121(96) dataset: Genuine Document Detection. The names of the vendors/machines were anonymised with capital letters.

personal experience of the officer in relation to real cases of false documents may strengthen the capacity of the officer to accurately detect document fraud. The level of training received, the length of the experience, or the specific type of operational experience in terms of first, second, or third line deployment did not significantly explain differences in performance in the time restricted scenario but may have played a role when time pressure was eliminated.

4.2. Machines

The evaluation of the performance of the machines was done on Test121(96) dataset. Initially, the results returned by the machines (without time constraint) were annotated by two master testers, where one of them did the inspections twice. Interestingly, high degree of discrepancy between annotations, i.e., κ values (kappa coefficient) [24] ranging from 0.521 to 0.80 for the different machines, was observed. This indicates the complexity of the task of interpreting the results of the machines, which constitutes a serious vulnerability per se in terms of security. The disagreements between the master testers might have resulted from both subjective interpretation by humans and inconsistency of the machines (e.g., due to technical failures). For the sake of evaluating the performance of the machines only the annotation prepared by the master tester, who went through all the documents/machines twice, was considered since it was deemed as more “reliable”.

Fig. 4 and 5 provide the basic scores for the false document and genuine document detection task resp.

As regards the task of detecting genuine documents the level of completeness (TPR) of the machines ranges from 50% (machine E) to 93.75% (machine G), whereas the level of completeness of the machines for detecting false documents (TNR) ranges from 52.1% (machine F) to 97.9% (machine E). The precision level scores for the machines in detecting genuine and false documents (PPV/NPV) oscillate between 73.7% (machine G) and 96% (machine E) for genuine documents and between 67.1% (machine E) and 92.9% (machine G) for false documents. The variation in precision (predictive value)

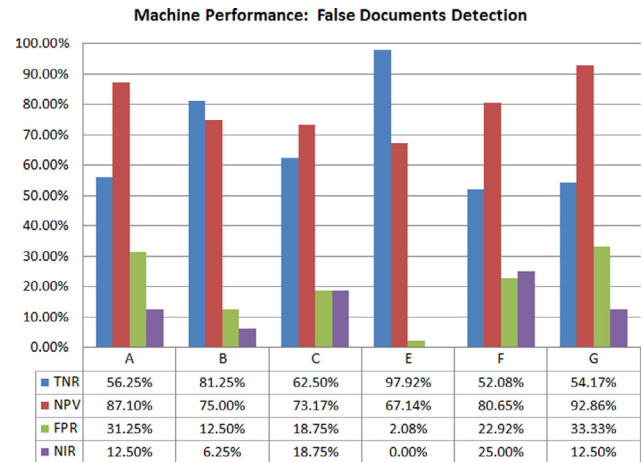


Fig. 5 – Machine Performance on Test121(96) dataset: False Document Detection. The names of the vendors/machines were anonymised with capital letters.

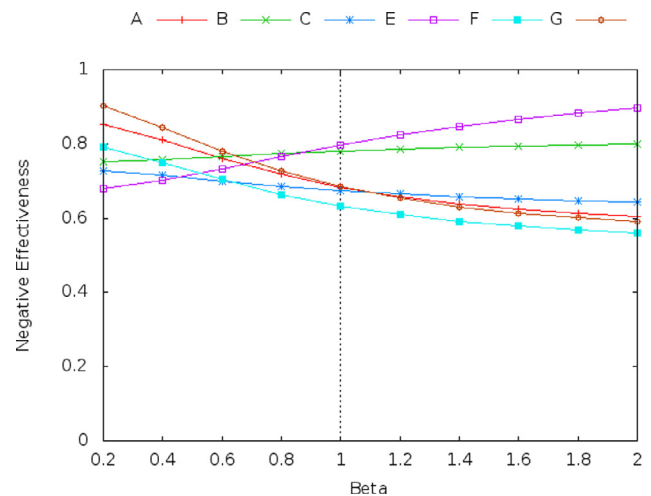


Fig. 6 – Negative Effectiveness: Machines.

across the different machines appears to be smaller than it is the case for completeness. Interestingly, the number of inconclusive answers for false documents (NIR) is on average higher than in the case of genuine documents (PIR).

Since from an operational perspective achieving high completeness might be more beneficial compared to achieving high precision, a comparison of Negative Effectiveness measures for the machines with different weightings (β coefficient) for the aforementioned characteristics is depicted in Fig. 6. In the case of detection of false documents, if equal weight is given to NPV (“precision”) and TNR (“completeness”) we can observe that machine E and B exhibit highest negative effectiveness (slightly below 0.8) in the entire machine pool, whereas machine F performs worst, i.e., negative effectiveness score is around 0.63. Giving higher weight to precision ($\beta < 1$) would result in machine G and A to score highest.

In order to understand the performance of the systems in a real “machine only” scenario in which only a binary classification is possible (in terms of genuine/green or

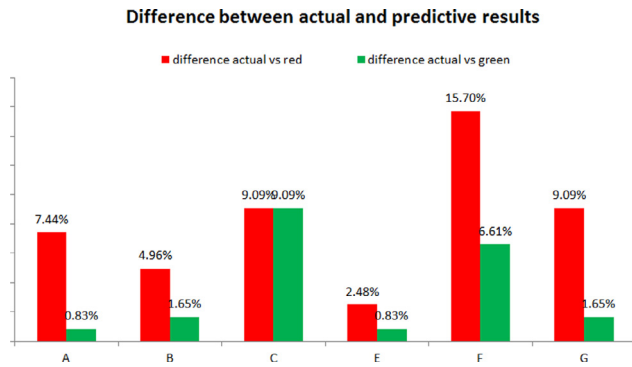


Fig. 7 – Difference between actual vs. predictive results – gain in accuracy vis-a-vis actual accuracy figures.

false/red), so-called predictive classification measures have been calculated. In the *predictive green* case, all inconclusive results have been interpreted as positive, while in the *predictive red* case, all indecision was automatically turned into rejection (e.g. an inconclusive document is a false document). Because Test121(96) is a balanced dataset, results are particularly significant. Accuracy improves for all machines in both green and red cases (see Fig. 7). However, apart for machine C, the red predictive case in which in case of indecision the document is flagged as suspicious yields a higher overall gain in accuracy. In particular, the improvement rate vis-a-vis the green predictive case ranges from 1.65% (E) to 9.09% (F).

Although providing facts on certain variability of the machines in terms of performing good in one aspect and bad in another (or to put it in other words, their complementarity) is not particularly interesting per se, we give them here to provide an approximation of the state of the art in automated document authenticity verification.

Looking again at the results from an operational perspective one could consider two important characteristics, namely, security and facilitation that could be compared and are often mentioned in the context of the entire border control processing chain. Here, in the context of evaluating the performance of the machines, “high” security could be interpreted as maximizing the number of false documents that are captured (high TNR score) and at the same time generating as few false acceptances as possible (high PPV score). On the other hand, “high” facilitation could be interpreted as maximizing the number of correctly classified genuine documents (high TPR score) while at the same time minimizing false rejections (high NPV rate). A summary of all rankings of the machines w.r.t. different scores is given in Table 3. Once again, it can be observed that there is no single machine that ranks highest in both security and facilitation. To be more precise, machine E ranks highest in terms of security (highest TNR and PPV), whereas machine G ranks highest in terms of facilitation (highest TPR and NPV). So we can conclude that there is no all-in-one solution as regards the pool of machines that were available for testing. As a matter of fact, the aforementioned machines, namely G and E, score highest in one of the categories (security vs. facilitation) and relatively low in the other, which might have resulted from certain response bias towards one of the choices. Hence, to answer the typical question of border control au-

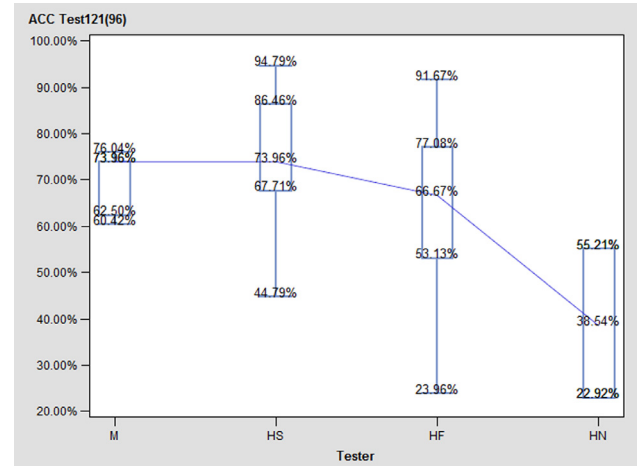


Fig. 8 – Comparison of Accuracy for Test121(96) dataset: M-machines, HS - document experts, HF-border guards (first line), HN-non-experienced testers.

thorities “what equipment is best?” there is no easy and clear answer. It depends on the specific characteristic one is interested in. If we consider that equipment that captures less than 60% of all false documents (A, G, F) or less than 60% of all genuine documents (C, E) is considered unacceptable (see equipment marked in red in Table 3) then the only remaining choice would be machine B, since all the other machines would be eliminated. Another possibility to consider an option would be to buy and deploy different equipment (i.e., more than one machine) and improve the performance through some kind of combining them into an ensemble of machines, where some heuristic could be used (e.g. majority voting), to determine the authenticity of the document. This, however, would require some further research, which could be pursued in future. In particular, carrying out exploration on the performance of such more sophisticated solution could be interesting in the context of second-line document authenticity verification due to different time constraints vis-a-vis first-line checks.

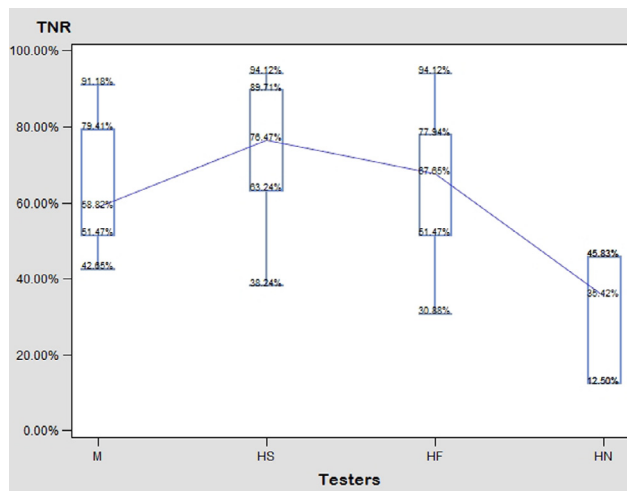
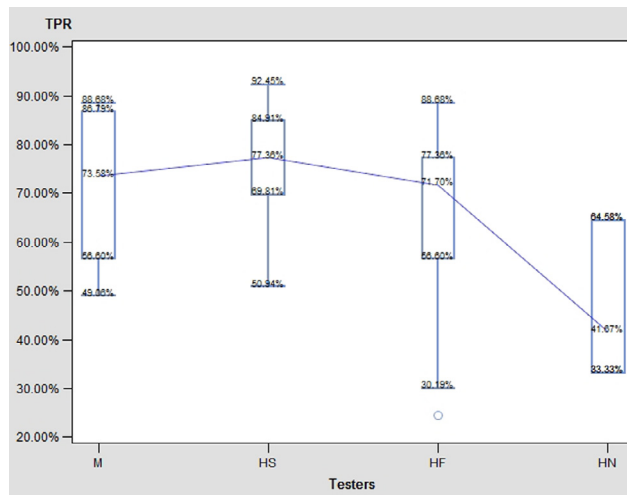
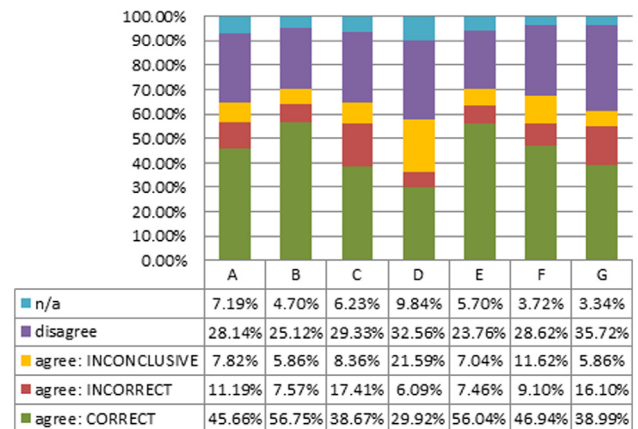
4.3. Humans versus machines

As regards the task of classifying documents into genuine/false the box-plot diagram in Fig. 8 depicts the respective accuracy performance on Test121(96) dataset. It can be observed that comparing the median values document experts and machines score highest (ca. 74,0%), followed by border control officers (66,7%) and non-experienced testers (38,5%). It has to be mentioned though that a vast majority of top-scoring document experts by far outperform the best DIS (94,8% score of the best document expert versus 76% score of the best DIS). Also some border guards outperform the best DIS, where the best observed score obtained by a border guard was 91,7%. The higher variability of human testers compared to the machines results is most likely due to the higher number of humans involved compared to the machines.

As regards the two other tasks, namely, the detection of false (genuine) documents a similar trend (to a large extent) could be observed for TNR (see Fig. 9) and TPR (see Fig. 10)

Table 3 – The summary of all rankings of the machines with respect to different scores. The equipment marked in red failed to detect 60% of the documents in the respective classes (genuine or false).

Rank	Classification	Detection of False Documents			Detection of Genuine Documents		
		Accuracy (balanced)	TNR completeness	NPV precision	Negative efficiency (balanced)	TPR completeness	PPV precision
1	B (76.0%)	E (97.9%)	G (92.9%)	E (79.7%)	G (93.8%)	E (96.0%)	G (82.6%)
2	G (74.0%)	B (81.3%)	A (87.1%)	B (78.0%)	A (89.6%)	B (85.0%)	A (81.4%)
3	E (74.0%)	C (62.5%)	F (80.7%)	G (68.4%)	F (72.9%)	F (76.1%)	B (77.3%)
4	A (72.9%)	A (56.3%)	B (75.0%)	A (68.4%)	B (70.8%)	C (75.7%)	F (74.5%)
5	F (62.6%)	G (54.1%)	C (72.2%)	C (67.4%)	C (58.3%)	A (74.1%)	C (65.9%)
6	C (60.4%)	F (52.1%)	E (67.1%)	F (63.3%)	E (50.0%)	G (73.8%)	E (65.8%)

**Fig. 9 – Comparison of True Negative Rate for Test121(96) dataset: M - machines, HS - document experts, HF - border guards (first line), HN - non-experienced testers.****Fig. 10 – Comparison of True Positive Rate for Test121(96) dataset: M - machines, HS - document experts, HF - border guards (first line), HN - non-experienced testers.****Analysis of Agreement between Humans and Machines****Fig. 11 – Analysis of Agreements on Test121 dataset: Humans vs. Machines.**

scores. In particular, in case of detection of false documents, comparing the median values both document experts and border control officers score higher than machines.

4.4. Humans with Machines

The aim of this part of the exercise was to get insight into the performance of the machines when assisted by humans, who make the final decision on document authenticity. The testers were asked to inspect a randomly selected limited set of documents from Test121 dataset with all the machines and to provide: (a) information whether they agree/disagree with the machine, and (b) their final decision. There was no time constraint in the context of this exercise. The total number of document inspections was 2673, where on average circa 8% of the documents from the entire Test121 dataset were tested on a given machine by all testers.

In general, it turned out that the level of agreement between the machines and humans oscillates around 58–70%. A fine-grained picture of the agreements and disagreements between humans and the machines is given in Fig. 11. In particular, the green and red bars correspond to agreements on correct (agree: CORRECT) and incorrect (agree: INCORRECT) decisions respectively. In particular, the latter group of agree-

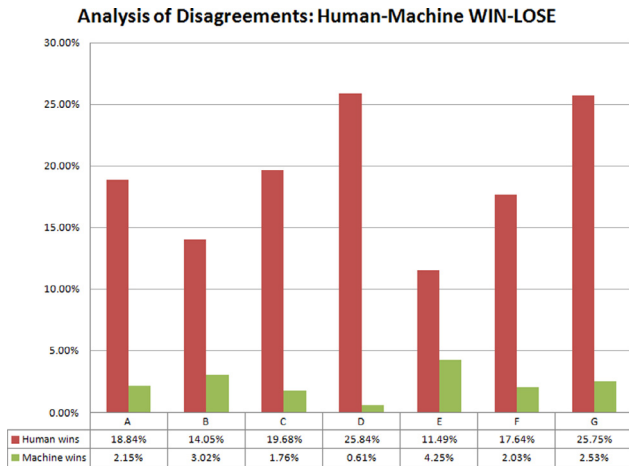


Fig. 12 – Analysis of Disagreements on Test121 dataset: Humans vs. Machines. The figures reflect the fraction of respective “wins” versus all cases (agreements and disagreements).

ments indicates an important vulnerability since both humans and machines failed to classify the document correctly.

The cases in which both humans and machines return an inconclusive answer are shown in yellow³. The level of disagreements oscillates around 23–35%, whereas the level of agreement ranges between 57.60% (D) and 70.18%–70.54% (B and E, respectively) depending on the machine.

For the sake of the analysis of disagreements we distinguish between two cases: (a) human and machine classify the document as genuine or false, (b) all other cases, e.g., either human or the machine is uncertain (inconclusive) or data is missing. Since the latter cases account only for 1–4% (depending on the machine) of all cases (i.e., total number of all agreements and disagreements) we focus here only on the disagreements of type (a). In particular, Fig. 12 provides the comparison of figures reflecting the percentage of disagreement cases in which humans (accompanied by a machine) and unassisted machines “win” resp. It can be observed that the cases of disagreements in which humans “won” range from 11 to 26%, whereas the disagreements in which machines return a correct answer account only for 0.5% to 4% of all cases. When only the disagreements cases are considered, humans win in 87.9% of them. These observations strengthen our hypothesis that machines alone perform worse than machines assisted by humans, who make final decisions. Interestingly, in case of non-experienced testers assisting the machines, the overall accuracy deteriorates by 7.14%.

4.5. Comparative error analysis for human vs machine

Fig. 13 shows the distribution of the Test121 documents in terms of difficulty level for both humans and machines. The level of difficulty⁴ was calculated based on how many ma-

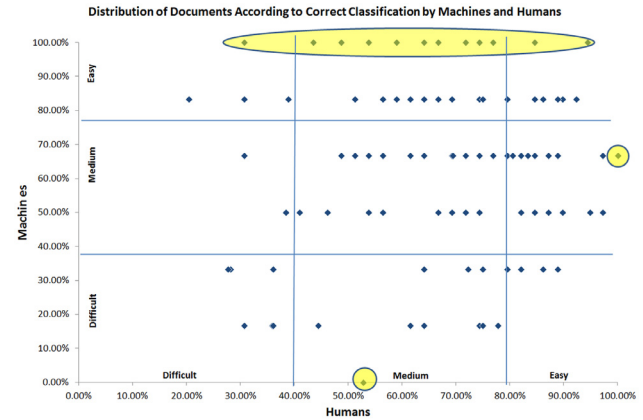


Fig. 13 – Distribution of documents according to difficulty level: Humans vs. Machines. The yellow circles refer to the documents considered by the document experts as difficult prior to carrying out the experiments (see Section 3.1).

chines/humans correctly classify the document as genuine or false. For example, if all the machines/humans (100%), correctly classify the document, the document is then considered as easy to detect. More precisely, if less than 39% classifications are successful, the document is considered difficult. If the success rate is within the range of 39–80% the difficulty level is considered to be medium, whereas success rates over 80% correspond to easy documents. Without engaging here in a forensic analysis of the documents that have been classified as easy or difficult to detect, the following general observations could be made:

- Documents with incorrect Machine Readable Zone (MRZ) format or check digits are very easy for the machines to detect; Humans have more problems in correctly calculating the check digits under time pressure;
- Machines find it relatively difficult to correctly identify counterfeit documents and page substitution. This could be because light exposure varies between Machines. This affects the quality of the Ultra Violet, Infra Red, and Visible Light images captured and the performance of the pattern matching algorithm.
- Humans tend to deal with exceptions better than machines (i.e. non-ICAO compliant documents, ‘famous’ type of forgeries object of several alerts at the EU level).
- At least 6 documents in the data-set proved to be very difficult for both humans and machines. 3 were genuine documents containing exceptions, production errors, or in very poor condition. The 3 forgeries had incorrect printing techniques. In general, the detection of abnormal printing techniques is very difficult for the machines.

5. Conclusions

The verification of the authenticity of a travel document is not a trivial task. Time constraints and the multi-tasking required in the first line of border checks as well as the growing complexity of the security of modern travel documents affect the effectiveness of document inspection capabilities. This article

³ The high number of inconclusives for machine D is due to its specifics of not providing a red/green answer.

⁴ Please note that this definition of difficulty level is different from the one introduced in Section 3.1, which reflects document expert opinion prior to carrying out the experiments.

reported on the Document Challenge II, the second exercise of this sort performed in Lisbon in 2013, whose aim was to evaluate the performance of humans, machines and their interaction in the task of verification of the authenticity of travel documents. The main findings of the Document Challenge II exercise are as follows:

- *Human only:*
 - on average humans are 65% accurate in the classification of documents as genuine or false,
 - non experienced testers are not better than chance (50:50),
 - as expected, humans perform better in the time-unrestricted scenario thus confirming that time pressure plays a role,
 - on average humans performed better on the genuine document detection task than on the false document detection,
 - document experts are better than border guards in both coverage and precision, although some exceptions/outliers are observed,
 - among the humans, a number of document experts could be identified with accuracy rates above 90% and very low false acceptance (FNR = 0 in one case) and false rejection (FPR = 2.08%). We hypothesize that their performance might have resulted from long-term experience and/or motivation, which could be studied in future research.
- *Machine only:*
 - interpretation of unclear results displayed by the DIS poses a serious vulnerability. This highlights the need to analyse the effectiveness of the user interface,
 - analogously, observed inconsistent DIS responses (same machine returning different results for the same document) which might have been caused by technical problems indicates the need to introduce additional evaluation metrics for measuring DIS response consistency,
 - no single machine is best (achieves the highest scores) at all “aspects” (e.g., security vs. facilitation). Only one machine among those tested balances the two detection tasks at the cost of a relative increase of corresponding errors when compared to the two best performing in false document detection and genuine document detection respectively,
 - the systems tested tended to have a higher indecision rate when inspecting false documents. In a fully automated scenario with a binary decision output, the handling of indecision becomes crucial
- *Machines assisted by Humans:*
 - the percentage of agreements between humans and machines ranges between 58–70% with only 29–56% agreement on correct classification,
 - in case of disagreements, humans tend to win against the machine. However the non-experienced humans are better off at always accepting the machine result because when they do not, they tend to make mistakes

Overall, the analysis of the performance of document inspection capabilities has shown that detection and clas-

sification performance is heterogeneous and depends on a number of factors. In general, experienced and trained border control officers outperformed the machines by far. The deployment of automated systems alone or with non-experienced/untrained officers represents an important vulnerability for the detection of false documents and should be studied and analysed thoroughly. Alone the interpretation of machine results constitutes a serious vulnerability.

Although the setup of the experiments carried out in the context of Document Challenge II in terms of the size of the test data, the number of machines, human experts and tests conducted allows to draw meaningful conclusions there are some obvious limitations when interpreting the reported results and analysis thereof. Firstly, the evaluation was carried out in a lab environment and not all factors that do impact decision making were taken into account. Secondly, although the selection of machines and documents used represent the state-of-the-art technology available on the market and known phenomena in terms of document fraud respectively the reported figures and findings can be considered only as an approximation of a full picture or more fine-grained findings one could obtain through exploring the performance of a broader range of document readers and increasing the number of human experts involved and documents tested (which is not an easy task due to administrative reasons) in the experiments. Finally, the test data reflects only “known” document fraud phenomena as of the time of carrying out the evaluations in 2013. The validity of the results in terms of performance of the machines after such a long lapse of time is unlikely: on the one hand, developments and improvements both in hardware and software may result in better performance rates for known falsifications; on the other, new technologies and methods may be used to falsify documents so that it is impossible to say whether the state of performance in this experiment is still valid. However, we believe the evaluation methodology and experiment setup used in Document Challenge II might constitute a good basis for planning suchlike regular explorations in future. To our best knowledge the reported results constitute the first published evaluation aiming at comparison of the performance of humans, machines and their interaction in the task of travel document authenticity validation done at such a scale.

Research in the area of usability, human-machine interaction and decision-making on the one hand as well as on the technical improvement of the performance of automated systems is needed in order to develop effective human-machine cooperation and increase the accuracy of document inspection capabilities at the border. Future simulations or assessments in close-to-real operational theatres (including variables such as human behaviour of passengers, multi-tasking, utilisation of additional information sources in the decision-making process, additional time-constraints not considered in our study, etc.) as well as experimental testing of different human-machine combinations using representative real travel documents (genuine and false) could contribute to a better understanding of the intrinsic vulnerabilities of human and technical document inspection capabilities and an overall improvement of false document detection at the border.

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